Progress Check

1. What is the ratio of volume to surface area of a sphere?
2. The relationship between the height and radius of the hemisphere is _____.
3. The volume of a sphere is the product of its surface area and _____.

7.3.8 Volume of frustum of a cone

Let H and h be the height of cone and frustum respectively, L and l be the slant height of the same.

If $R,\ r$ are the radii of the circular bases of the frustum, then volume of the frustum of the cone is the difference of the volumes of the two cones.

$$V = \frac{1}{3}\pi R^2 H - \frac{1}{3}\pi r^2 (H - h)$$

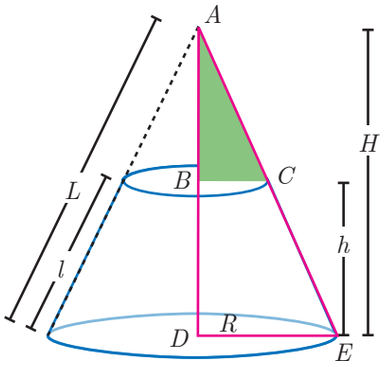


Fig. 7.34

Since the triangles ABC and ADE are similar, the ratio of their corresponding sides are proportional.

Therefore, $\frac{H-h}{H} = \frac{r}{R} \Rightarrow H = \frac{hR}{R-r}$ ...(1)

$$V = \frac{1}{3}\pi R^2 H - \frac{1}{3}\pi r^2 (H - h)$$

$$= \frac{\pi}{3} H(R^2 - r^2) + \frac{1}{3}\pi r^2 h$$

$$= \frac{\pi}{3}\frac{hR}{R-r}(R^2 - r^2) + \frac{\pi}{3} r^2 h \quad \text{[using (1)]}$$

$$= \frac{\pi}{3} hR(R + r) + \frac{\pi}{3} r^2 h$$

Thinking Corner

Is it possible to obtain the volume of the full cone when the volume of the frustum is known?

Volume of a frustum $= \frac{\pi h}{3}(R^2 + Rr + r^2)$ cu. units

Example 7.23 If the radii of the circular ends of a frustum which is 45 cm high are 28 cm and 7 cm, find the volume of the frustum.

Solution Let $h,\ r$ and R be the height, top and bottom radii of the frustum.

Given that, $h = 45$ cm, $R = 28$ cm, $r = 7$ cm

$$\text{Volume} = \frac{1}{3}\pi\left[R^2 + Rr + r^2\right]h \text{ cu. units}$$

$$= \frac{1}{3} \times \frac{22}{7} \times \left[28^2 + (28 \times 7) + 7^2\right] \times 45$$

$$= \frac{1}{3} \times \frac{22}{7} \times 1029 \times 45 = 48510$$

Therefore, volume of the frustum is 48510 cm^3

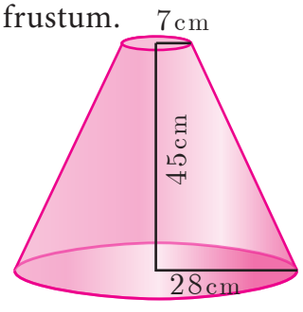


Fig. 7.35

7 Mensuration.indd 289 16-12-2020 03:18:07 PM